DETAIL SPECIFICATION SHEET

ARC FAULT PROTECTION SYSTEM, CONTINUOUS THERMAL MONITORING DEVICE, THERMAL IONIZATION TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-32656.

REQUIREMENTS:

Continuous thermal monitoring device (CTMD) performance characteristics. The thermal ionization detector (TID) CTMD is a type that shall respond to small particles of insulation emitted by overheated cables or from coated junctions and generate an alarm signal. If this signal persists (approximately 4 seconds), the Arc Fault Protection System (AFPS) control unit shall alarm.

Components. The TID CTMD shall consist of an ionization chamber with test electrode and electrical components to provide for signal processing and transfer within an electromagnetic interference shielded enclosure.

Envelope. The TID CTMD, without respect to connecting hardware and cabling, shall occupy an envelope of no more than 6 centimeters (2.5 inches) wide by 6 centimeters (2.5 inches) high by 4 centimeters (1.5 inches) diameter.

Clean air operation. In a clean air environment in which the air quality is maintained to be free of smoke, dust, and fumes, and the built-in test (BIT) is not activated, the output signal voltage range shall be between 56 percent and 70 percent of the ionization chamber supply voltage. Also, the output signal shall not fall below 46 percent of the chamber operating voltage for more than 3.5 seconds within a 1-minute test time.

BIT command. The CTMD shall incorporate a performance test circuit that shall permit electrical testing of the ionization chamber. Activation of the BIT electrode shall produce an output voltage of less than 38 percent of the chamber supply voltage when the BIT has been activated for at least 20 seconds. An alarm signal shall be defined to be an output voltage equal to or less than 46 percent of the chamber supply voltage.

EXAMINATIONS AND TESTS:

Voltage outputs. The CTMD shall be tested with the manufacturer's working voltage applied. The required voltage outputs of the clean air and BIT response shall be verified to not fall below 46 percent of the chamber operating voltage for more than 3.5 seconds within a 1-minute test time.

Sensitivity. For first article tests only, when the thermal detector detects off-gassing from overheated insulation, the device shall be subjected to the sensitivity test in accordance with UL 268 with the following exceptions:

a. The alarm signal shall activate within the limits specified in table I.

b. The requirement for the directionality test in accordance with UL 268 shall be omitted.

TABLE I. Visible smoke obscuration limits (gray smoke).

Minimum obscuration	0.5 percent per foot
Maximum obscuration	4.0 percent per foot

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